Modeling and Forecasting Exchange Rate Dynamics Between CAD and AD

EC6011 Group Project – MSc Business Analytics 2024/25

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We hereby declare that this work is entirely our own and has not been submitted as part of any other examination or assignment. Any use of the work of others in this assignment is duly acknowledged.

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Introduction

In an increasingly interconnected global economy, understanding how exchange rates react to domestic and international economic forces is crucial for policymakers and businesses. To better understand this interplay, we will analyse and forecast the exchange rate dynamics between Canada and Australia, with Canada serving as the home country and Australia as the foreign country. Our analysis is based on a robust dataset collected over ten years every quarter. The data comprises four key variables: the nominal exchange rate between the Canadian (CAD) and Australian dollars (AD), the derived real exchange rate, and the Consumer Price Index (CPI) for both Canada and Australia.

To facilitate a meaningful comparison, all series will be transformed using natural logarithms and, where necessary, normalized to a common base year (2014). This will allow us to capture relative changes and growth rates over time. We will then test the validity of Purchasing Power Parity (PPP) by examining both its absolute and relative forms, which will evaluate the equilibrium relationship between exchange rates and price levels across these two economies.

Finally, we will employ Box-Jenkins models such as Auto Regressive (AR), Moving Average (MA), Auto Regressive Moving Average (ARMA), Auto Regressive integrated Moving Average (ARIMA) to forecast the real exchange rate, providing insights into its dynamic behaviour.

Rationale behind country selection

In selecting Canada and Australia for our analysis, we aimed to examine two countries with relatively stable currencies and sufficient and reliable economic data. As the home country, Canada is characterized by a robust financial system and prudent monetary policies. Australia, our chosen foreign country, offers a complementary perspective with a stable commodity-

driven economy. Both nations provide extensive historical data on key economic indicators for testing exchange rate theories and purchasing power parity.

Data Description and Methodology

Data Collection

Data for this study were obtained from highly reputable sources to ensure our analysis's validity and reliability. The nominal exchange rate between Canada (home) and Australia (foreign) was sourced from OFX, a reputable Australian fintech company. Meanwhile, official Consumer Price Index (CPI) data was obtained from Statistics Canada for the Canadian economy and the Australian Bureau of Statistics for Australia. We selected a ten-year sample period from 2014 to 2024, drawing upon quarterly observations.

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Table 1: Initial dataset

Quarter	CPI_Canada	CPI_Australia	Nominal_ExRate
Q1 2014	124.00	105.40	1.01
Q2 2014	125.63	105.90	0.98
Q3 2014	125.73	106.40	0.99
Q4 2014	125.27	106.60	1.03
Q1 2015	125.33	106.80	1.02
Q2 2015	126.77	107.50	1.05
Q3 2015	127.23	108.00	1.05
Q4 2015	126.93	108.40	1.04
Q1 2016	127.27	108.20	1.01
Q2 2016	128.73	108.60	1.04
Q3 2016	128.80	109.40	1.01
Q4 2016	128.70	110.00	1.00

Data Preparation

We consolidated our data in Excel by converting monthly observations into quarterly data. For each quarter (Q1, Q2, Q3, Q4), we calculated the average of the constituent months over the ten-year period. The consolidated file was then saved as a CSV and imported into Jupyter Notebook. Once in Python, we implemented a function to convert the date column into a consistent "yyyy-mm-dd" format reflecting quarterly dates, ensuring that the data were appropriately formatted for subsequent analysis.

Date	CPI_Canada	CPI_Australia	Nominal_ExRate
2014-03-01	124.00	105.4	1.01
2014-06-01	125.63	105.9	0.98
2014-09-01	125.73	106.4	0.99
2014-12-01	125.27	106.6	1.03
2015-03-01	125.33	106.8	1.02
2015-06-01	126.77	107.5	1.05
2015-09-01	127.23	108.0	1.05
2015-12-01	126.93	108.4	1.04
2016-03-01	127.27	108.2	1.01

Figure 1: Dataset with converted date format

Calculation of the Real Exchange Rate

Next, we calculated the real exchange rate based on the nominal exchange rate and CPI values we sourced. We used the formula recommended by Kıpıcı & Kesriyeli (1997) to calculate the real exchange rate (RER) from our nominal exchange rate and CPI datasets. This approach is expressed as R = eP*/P, where e represents the nominal exchange rate, P* is the foreign country's CPI, and P is the home country's CPI.

Date	CPI_Canada	CPI_Australia	Nominal_ExRate	Real_ExRate
2014-03-01	124.00	105.4	1.01	0.86
2014-06-01	125.63	105.9	0.98	0.83
2014-09-01	125.73	106.4	0.99	0.84
2014-12-01	125.27	106.6	1.03	0.88
2015-03-01	125.33	106.8	1.02	0.87
2015-06-01	126.77	107.5	1.05	0.89
2015-09-01	127.23	108.0	1.05	0.89

Figure 2: Dataset with computed real exchange rate column

Re-basing the CPI

We re-based both CPIs to a common base of 100 in 2014, which enabled us to directly compare the relative changes in price levels over time between Canada and Australia. Normalizing the data removed differences in scale and highlighted percentage changes from the base year, thereby making trends more visually interpretable.

Date	CPI_Canada	CPI_Australia	Nominal_ExRate	Real_ExRate	Normalized_CPI_Canada	Normalized_CPI_Australia
2014-03-01	124.00	105.4	1.01	0.86	100.000000	100.000000
2014-06-01	125.63	105.9	0.98	0.83	101.314516	100.474383
2014-09-01	125.73	106.4	0.99	0.84	101.395161	100.948767
2014-12-01	125.27	106.6	1.03	0.88	101.024194	101.138520
2015-03-01	125.33	106.8	1.02	0.87	101.072581	101.328273

Figure 3: Dataset with Normalized CPI columns computed

Log transformation

We converted the nominal exchange rate, real exchange rate, and CPIs for both Canada and Australia into their logarithmic forms, primarily to stabilize variance and manage scale differences.

Date	CPI_Canada	CPI_Australia	Nominal_ExRate	Real_ExRate	Normalized_CPI_Canada	Normalized_CPI_Australia	log_Nominal_ExRate	log_Real_ExRate
2014-03-01	124.00	105.4	1.01	0.86	100.000000	100.000000	0.009950	-0.150823
2014-06-01	125.63	105.9	0.98	0.83	101.314516	100.474383	-0.020203	-0.186330
2014-09-01	125.73	106.4	0.99	0.84	101.395161	100.948767	-0.010050	-0.174353
2014-12-01	125.27	106.6	1.03	0.88	101.024194	101.138520	0.029559	-0.127833
2015-03-01	125.33	106.8	1.02	0.87	101.072581	101.328273	0.019803	-0.139262

log_CPI_Canada	log_CPI_Australia
4.820282	4.657763
4.833341	4.662495
4.834137	4.667206
4.830471	4.669084
4.830950	4.670958

Figure 4: Converted logarithms of all variables

Exploratory Data Analysis

We plotted a time series graph for each variable and noted the following observations:

Nominal Exchange rate

The graph reveals fluctuations over the decade, with periods where the rate increases. Implying a stronger Canadian dollar relative to the Australian dollar, while periods where it decreases suggest a relative weakening of the Canadian dollar. These variations can be attributed to changes in economic conditions, monetary policies, and market dynamics influencing the currency values. Overall, the exchange rate remains within a moderate band.

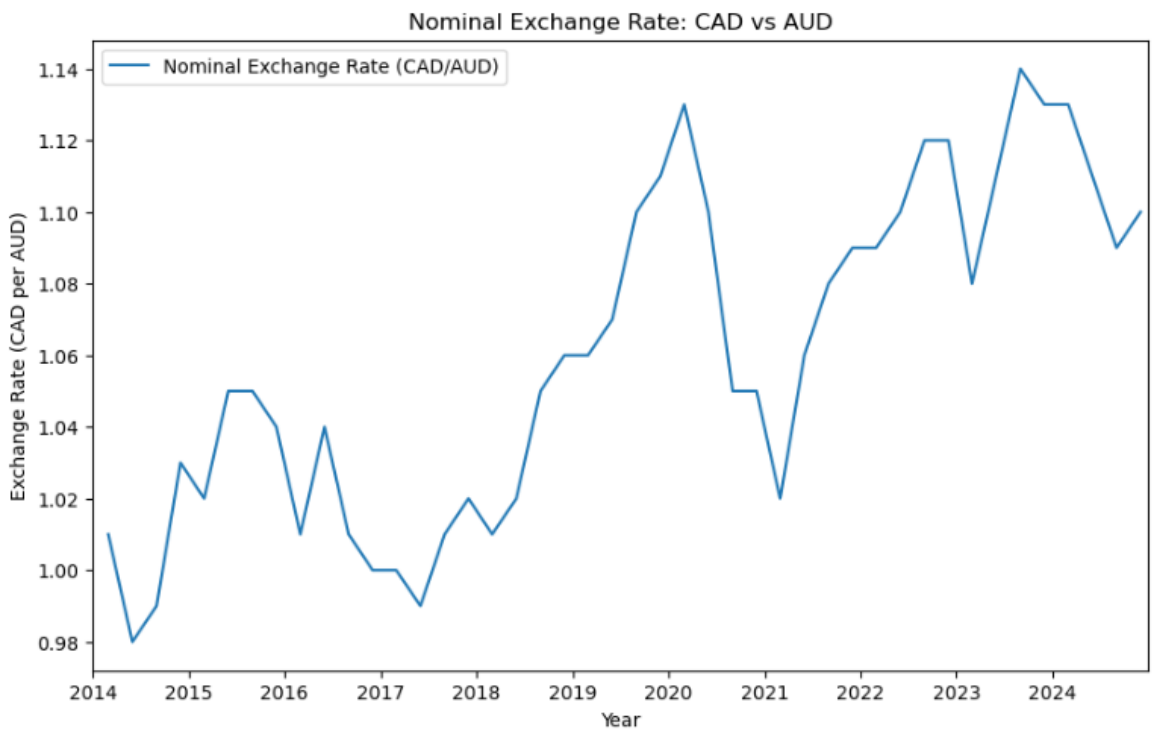

Figure 5: Nominal exchange rate between the Canadian dollar and the Australian dollar from 2014 to 2024

Real Exchange Rate

This chart illustrates the real exchange rate between Canada and Australia over the 2014–2024 period, showing how inflation-adjusted currency values have evolved. From a relatively lower

level in 2014, the rate rose significantly around 2020, suggesting that Canada's real value strengthened to Australia's before experiencing some downward adjustments.

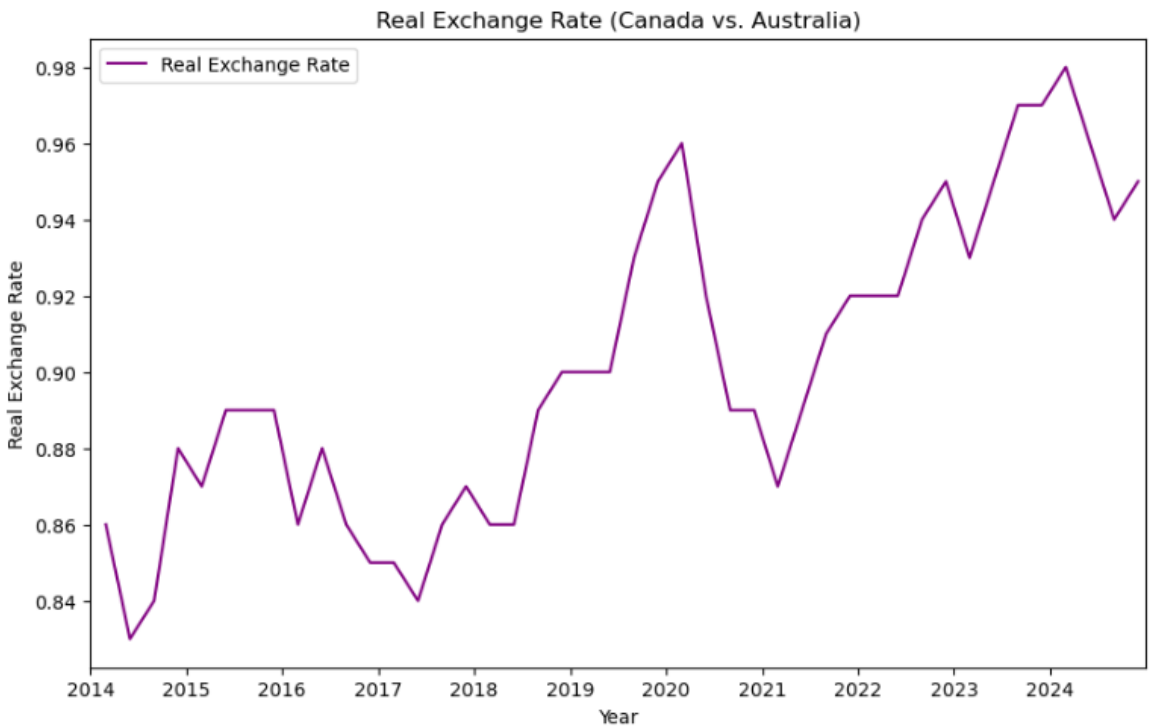

Figure 6: Real exchange rate between the Canadian dollar and the Australian dollar from 2014 to 2024

Normalized CPI Comparison

The following chart compares Canada's and Australia's CPIs, both re-based to 100 in 2014, to highlight relative changes over time. However, both countries show a clear upward trend. Australia's CPI exhibits a slightly steeper climb in certain periods, while Canada's rise appears more gradual. Despite temporary dips and variations, both indices converge toward similar levels by 2024.

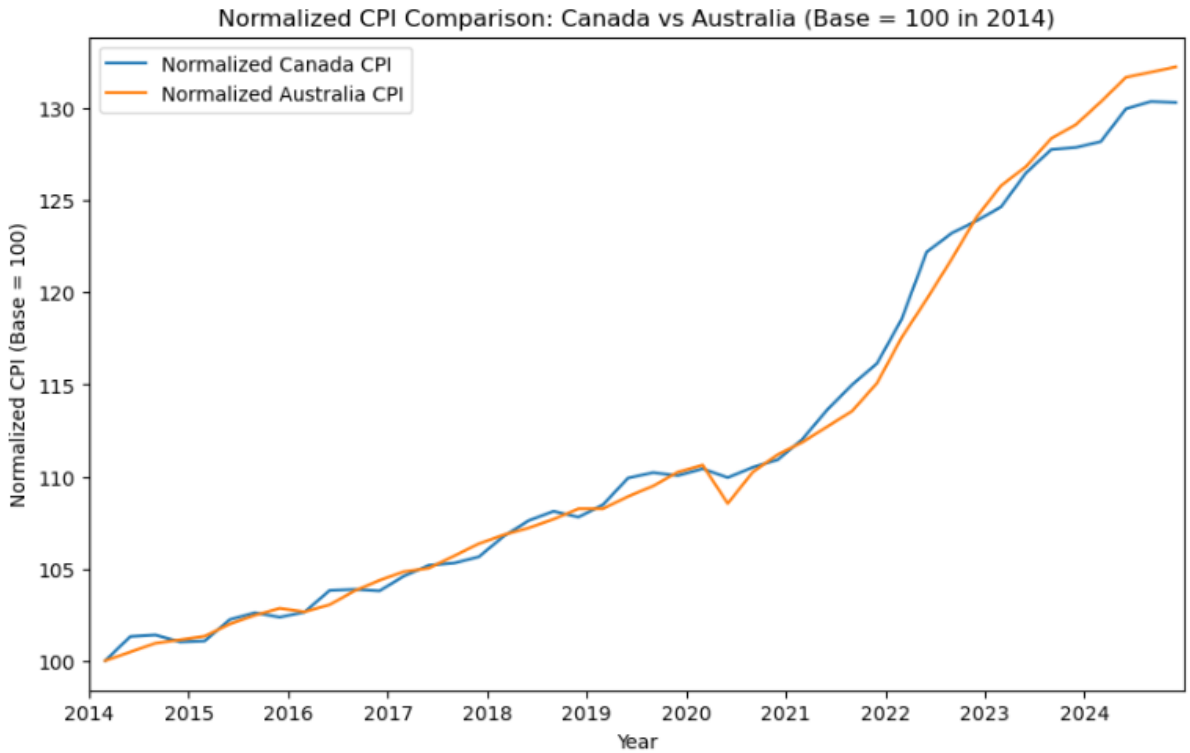

Figure 7: Re-based CPI comparison

Log transformation

The log of nominal and real exchange rates fluctuates over time, indicating periodic rises and falls in currency values once scaled by inflation. In contrast, the log of CPI for both Canada and Australia shows a clear upward trend, reflecting persistent inflation across the decade.

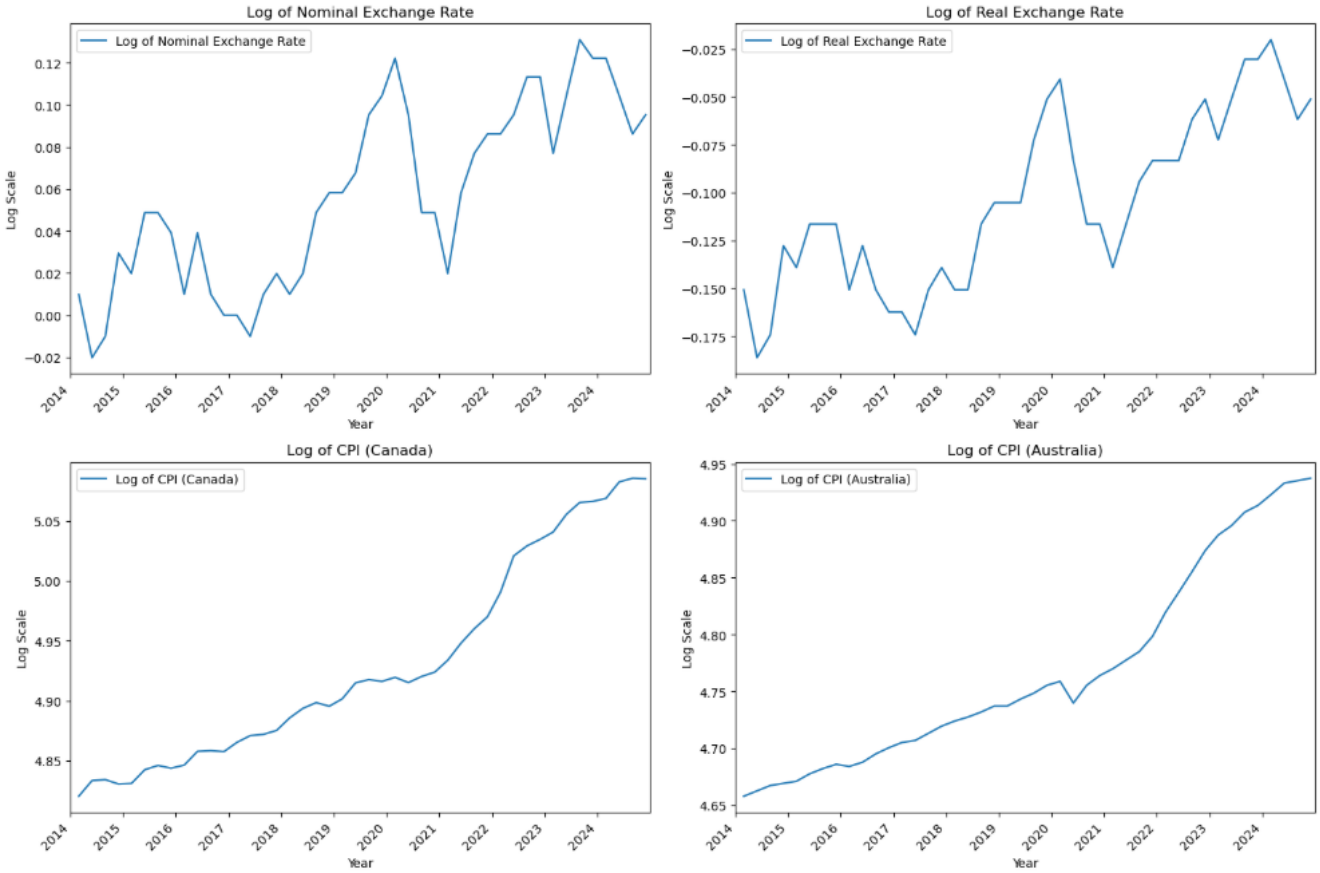

Figure 8: Visualizing the log transformations of all variables

Stationarity Test

We used the Augmented Dickey-Fuller (ADF) test to check whether the statistical properties of our time series data remain consistent over time, to ensure it is stationary (Assaf, 2006). The initial ADF test indicated that the original data is non-stationary. From the ADF statistics and p-values, neither the CPI series nor the nominal exchange rate appear to be stationary in levels. Similarly, the real exchange rate shows non-stationary behaviour. This indicates that their means and variances change over time, suggesting a need for differencing (Assaf, 2006). Thus, we fail to reject the null hypothesis of non-stationarity data which also indicated the time series data has a trend and other features of non-stationarity.

```
ADF Test for CPI_Canada:
ADF Statistic: 3.1422
p-value: 1.0000
Critical Values: {'1%': -3.6461350877925254, '5%': -2.954126991123355, '10%': -2.6159676124885216}
Conclusion: Not Stationary

ADF Test for CPI_australia:
ADF Statistic: 1.4196
p-value: 0.9972
Critical Values: {'1%': -3.596635636000432, '5%': -2.933297331821618, '10%': -2.6049909750566895}
Conclusion: Not Stationary

ADF Test for Nominal exchange rate:
ADF Statistic: -1.6774
p-value: 0.4428
Critical Values: {'1%': -3.5925042342183704, '5%': -2.931549768951162, '10%': -2.60406594375338}
Conclusion: Not Stationary

ADF Test for Real exchange rate:
ADF Statistic: -1.5601
p-value: 0.5036
Critical Values: {'1%': -3.5925042342183704, '5%': -2.931549768951162, '10%': -2.60406594375338}
Conclusion: Not Stationary

ADF Test for log_nominal_exchange_rate:
ADF Statistic: -1.6787
p-value: 0.4421
Critical Values: {'1%': -3.5925042342183704, '5%': -2.931549768951162, '10%': -2.60406594375338}
Conclusion: Not Stationary

ADF Test for log_real_exchange_rate:
ADF Statistic: -1.5719
p-value: 0.4977
Critical Values: {'1%': -3.5925042342183704, '5%': -2.931549768951162, '10%': -2.60406594375338}
Conclusion: Not Stationary

ADF Test for log_CPI_Canada:
ADF Statistic: 3.0483
p-value: 1.0000
Critical Values: {'1%': -3.6461350877925254, '5%': -2.954126991123355, '10%': -2.6159676124885216}
Conclusion: Not Stationary

ADF Test for log_CPI_Australia:
ADF Statistic: 1.3273
p-value: 0.9968
Critical Values: {'1%': -3.596635636000432, '5%': -2.933297331821618, '10%': -2.6049909750566895}
Conclusion: Not Stationary
```

Figure 9: Results of ADF stationarity test

First Order Differencing

After taking the first difference of each variable, we applied the ADF test again to see if these differenced series became stationary. From this exercise, we observed stationarity in all variables except Canada's CPI and log transformation.

CPI_Canada has an ADF statistic of -2.2916 and a p-value of 0.1747, while log_CPI_Canada has an ADF statistic of –2.4079 and a p-value of 0.1395. In both cases, the ADF statistics are above the 5% critical value of –2.9371, and the p-values exceed 0.05, indicating non-stationarity.

```
ADF Test for CPI_Canada (Differenced):
ADF Statistic: -2.2916
p-value: 0.1747
Critical Values: {'1%': -3.6055648906249997, '5%': -2.937069375, '10%': -2.606985625}
Conclusion: Not Stationary

ADF Test for CPI_australia (Differenced):
ADF Statistic: -3.9642
p-value: 0.0016
Critical Values: {'1%': -3.596635636000432, '5%': -2.933297331821618, '10%': -2.6049909750566895}
Conclusion: Stationary

ADF Test for Nominal exchange rate (Differenced):
ADF Statistic: -6.5191
p-value: 0.0000
Critical Values: {'1%': -3.596635636000432, '5%': -2.933297331821618, '10%': -2.6049909750566895}
Conclusion: Stationary

ADF Test for Real exchange rate (Differenced):
ADF Statistic: -6.7662
p-value: 0.0000
Critical Values: {'1%': -3.596635636000432, '5%': -2.933297331821618, '10%': -2.6049909750566895}
Conclusion: Stationary

ADF Test for log_nominal_exchange_rate (Differenced):
ADF Statistic: -6.5938
p-value: 0.0000
Critical Values: {'1%': -3.596635636000432, '5%': -2.933297331821618, '10%': -2.6049909750566895}
Conclusion: Stationary

ADF Test for log_real_exchange_rate (Differenced):
ADF Statistic: -6.8463
p-value: 0.0000
Critical Values: {'1%': -3.596635636000432, '5%': -2.933297331821618, '10%': -2.6049909750566895}
Conclusion: Stationary

ADF Test for log_CPI_Canada (Differenced):
ADF Statistic: -2.4079
p-value: 0.1395
Critical Values: {'1%': -3.6055648906249997, '5%': -2.937069375, '10%': -2.606985625}
Conclusion: Not Stationary

ADF Test for log_CPI_Australia (Differenced):
ADF Statistic: -4.3171
p-value: 0.0004
Critical Values: {'1%': -3.596635636000432, '5%': -2.933297331821618, '10%': -2.6049909750566895}
Conclusion: Stationary
```

Figure 10: ADF testing results after first differencing

Second Order Differencing

After the second differencing, both variables were observed to be stationary. CPI_Canada_diff2 and log_CPI_Canada_diff2 show ADF test statistics of –3.7689 and –3.7145, both more negative than the 5% critical value of –2.9572. Their p-values 0.0036 and 0.0039 are also below 0.05, indicating we can reject the null hypothesis of non-stationarity and confirm stationarity.

```
ADF Test for CPI_Canada_diff2 (Second Differenced):
ADF Statistic: -3.7369
p-value: 0.0036
Critical Values: {'1%': -3.653519805908203, '5%': -2.9572185644531253, '10%': -2.6175881640625}
Conclusion: Stationary

ADF Test for log_CPI_Canada_diff2 (Second Differenced):
ADF Statistic: -3.7145
p-value: 0.0039
Critical Values: {'1%': -3.653519805908203, '5%': -2.9572185644531253, '10%': -2.6175881640625}
Conclusion: Stationary
```

Figure 11: ADF testing results after second differencing

Autocorrelation Function (ACF)

The Autocorrelation Function (ACF) plot below explains the significant spike at lag 1, which indicates a possible Moving Average (MA) component in the time series data, and the PACF plot explains the significance of lag 1which indicates a possible Autoregressive (AR) component in the time series data.

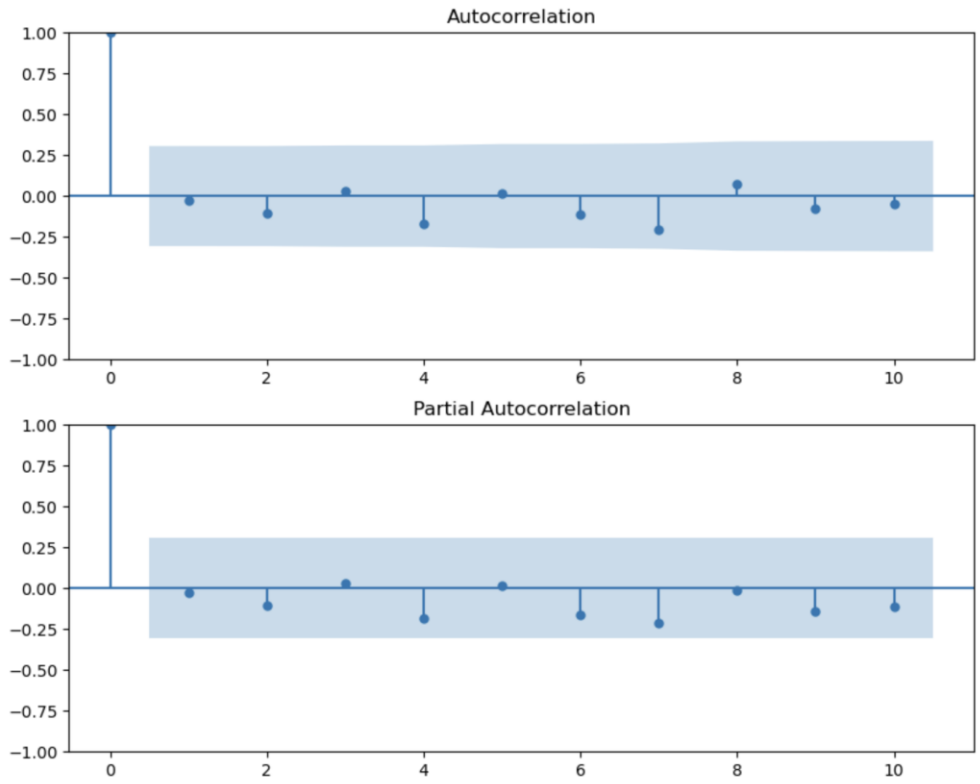

Figure 12: Autocorrelation and Partial Autocorrelation Functions

Purchase Power Parity

Purchasing Power Parity (PPP) is the theory that under ideal conditions, exchange rates adjust so that a standard selection of goods costs the same in different countries when expressed in a common currency (Dornbusch, 1985). In other words, differences in national price levels should be offset by changes in the exchange rate, ensuring equal purchasing power between Canada and Australia.

Test for validity of Absolute Purchase Power Parity

The absolute PPP is performed in the project to examine whether the exchange rate between Canada and Australia reflects the relative price levels of goods and services.

The graph below summarises the Absolute PPP between Canadian Dollar and Australian Dollar from the period of 2014 to 2024.

The ADF test statistics results is less negative than the critical values, p-value (0.5036) is higher than 0.05 critical level. Thus, we fail to reject the null hypothesis of non-stationarity. This further suggests that the real exchange rate is non-stationary which indicates Absolute PPP does not hold in the case.

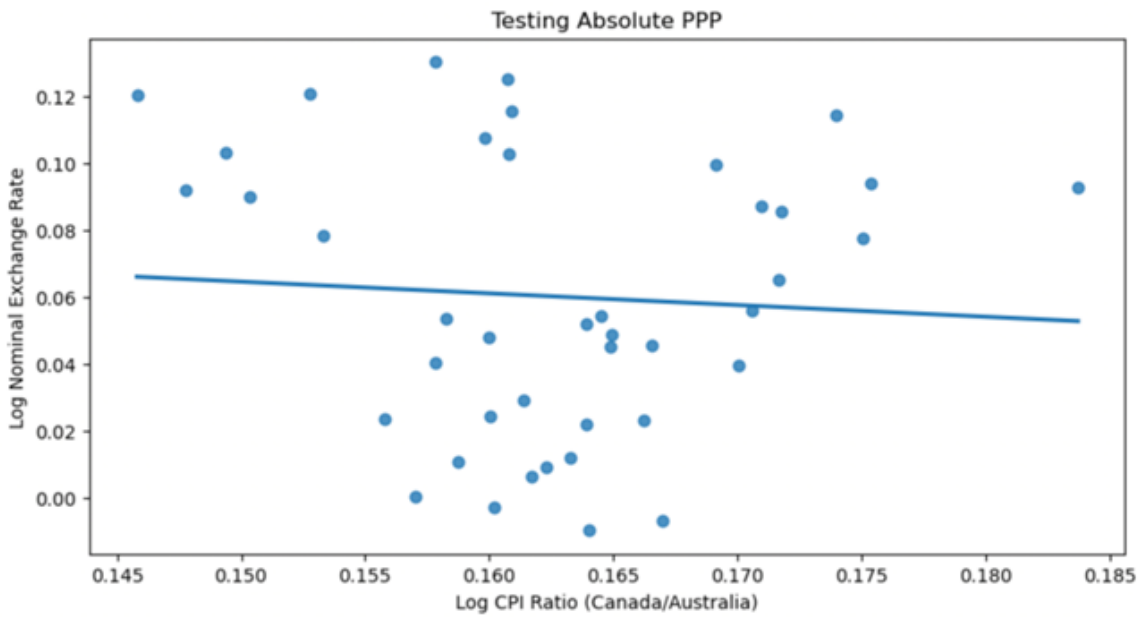

Figure 13: Scatter Plot for Testing Absolute PPP

```
Testing Absolute PPP (Stationarity of Real Exchange Rate):
ADF Statistic: -1.560103830738448
p-value: 0.5035974211094394
Critical Values: {'1%': -3.5925042342183704, '5%': -2.931549768951162, '10%': -2.60406594375338}
Conclusion: Not Stationary ❌ (PPP does not hold)
```

Figure 14: ADF Test for Real Exchange Rate and Absolute PPP

Test for validity of Relative Purchase Power Parity

Relative PPP is calculated to analyse the changes in the price levels between Australia and Canada as how they affect exchange rate movements over time.

The graph below summarises the Relative PPP between Canadian Dollar and Australian Dollar from the period of 2014 to 2024. The ADF test (-2.8421) is lower than the critical value, and the p-value (0.0525) is greater than the critical level 0.05, which indicates that inflation differences do not systematically determine the exchange rates. Hence, relative PPP holds.

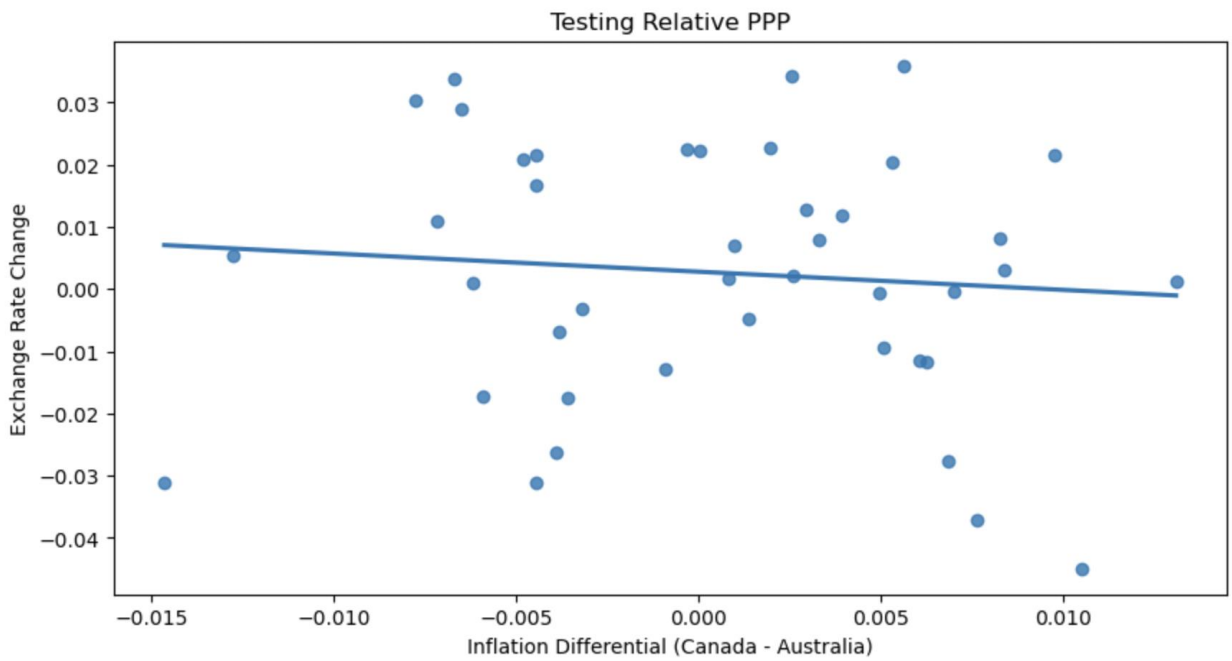

Figure 15: Scatter Plot for Testing Relative PPP

```
Testing Relative PPP (Stationarity of Inflation Differential & Exchange Rate Change):
ADF Statistic: -2.8944259281624993
p-value: 0.046000036384619664
Critical Values: {'1%': -3.6327426647230316, '5%': -2.9485102040816327, '10%': -2.6130173469387756}
Conclusion: Stationary ✅ (Relative PPP holds)
```

Figure 16: ADF Test for Relative PPP

The OLS regression was performed to test whether the inflation differential between Canada and Australia explains the changes in their exchange rate, as predicted by relative Purchasing Power Parity (PPP).

The ADF test suggests that the combination of exchange rate changes and the inflation differential forms a stationary series, implying a long-run relationship consistent with relative PPP. In contrast, the OLS regression finds that inflation differentials explain little of the short-run variation in exchange rate changes.

```
OLS Regression Results (Relative PPP Test):
                            OLS Regression Results
==============================================================================
Dep. Variable:     Exchange_Rate_Change   R-squared:                       0.008
Model:                              OLS   Adj. R-squared:                 -0.016
Method:                   Least Squares   F-statistic:                    0.3349
Date:                  Thu, 10 Apr 2025   Prob (F-statistic):              0.566
Time:                          09:39:57   Log-Likelihood:                 104.27
No. Observations:                    42   AIC:                            -204.5
Df Residuals:                        40   BIC:                            -201.1
Df Model:                             1
Covariance Type:              nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const              0.0028      0.003      0.871      0.389      -0.004       0.009
Inflation_Diff    -0.2922      0.505     -0.579      0.566      -1.312       0.728
==============================================================================
Omnibus:                        1.912   Durbin-Watson:                   1.987
Prob(Omnibus):                  0.384   Jarque-Bera (JB):                1.751
Skew:                          -0.398   Prob(JB):                        0.417
Kurtosis:                       2.395   Cond. No.                         158.
==============================================================================
```

Figure 17: OLS Regression for Relative PPP

Modelling and Forecasting

In the modelling phase using the Box-Jenkins methodology, our initial step involves ensuring the time series data is stationary. To achieve this, we will employ the Augmented Dickey-Fuller (ADF) test. We will then develop and estimate models such as Autoregressive (AR), Moving Average (MA), Autoregressive Moving Average (ARMA), and specific Autoregressive Integrated Moving Average (ARIMA) models like (1,1,0), (0,1,1), (1,1,1), (2,1,0), (0,1,2), and (2,1,2).

Autoregressive model (AR)

An autoregressive (AR) model is a time series model where the current observation is explained by its own previous values and a random error term (Shibata, 1976). AR model is denoted by (p).

Interpretation on ACF plot for AR model

The key observation on the ACF plot is the highlight on the initial (Lag 0 = 1.0); the lag is always 1 because it represents a perfect autocorrelation. The lag gradually decreases but remains significant until lag 5 before fading out, which indicates a slow decline of autocorrelation, which is a nature of the AR model. Since ACF gradually declines, it proves the nature of AR process. The blue shaded region represents the confidence interval, which describes the significant autocorrelation.

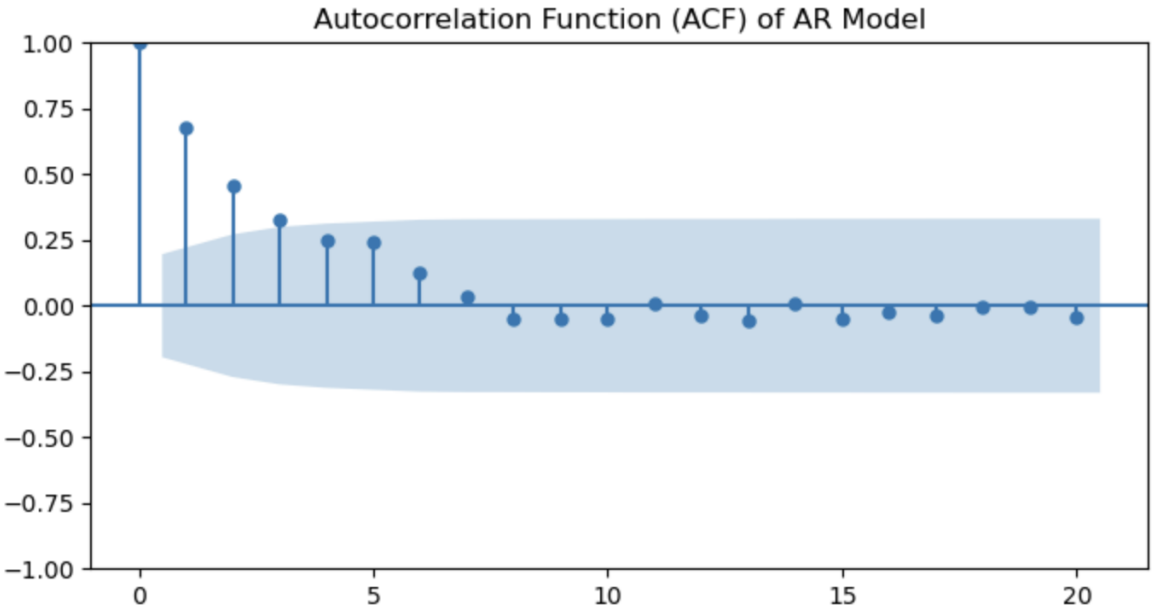

Figure 18: ACF plot for AR model

Interpretation on Partial Autocorrelation function (PACF) plot

In lag 1, we can observe a significant spike, which indicates that the real exchange rate at time (t) is strongly correlated with the past values (t-1), which is a key component of AR model. After the first lag, all the values start to descend within the shaded region, representing statistical insignificance, which indicates the past values beyond lag 1 do not add more predictive power, which further specifies a higher AR(p) model is required.

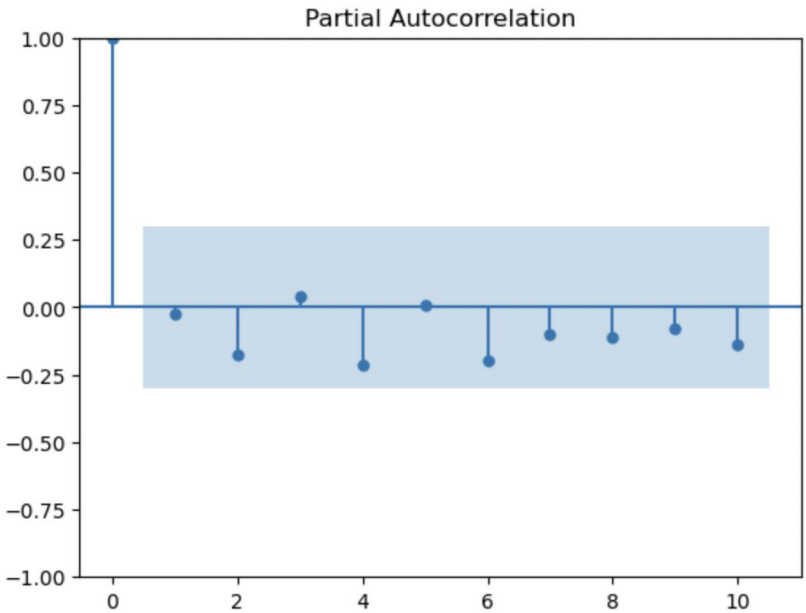

Figure 19: Partial Autocorrelation function (PACF) plot

AR model summary

The Ljung-Box test shows no significant autocorrelation in the residuals, which is p = 0.98, indicating the model captures the time series dependencies well. The Jarque-Bera test indicates the residuals follow a normal distribution with the p = 0.35 and no strong evidence of heteroskedasticity, which is p = 0.56, which confirms a stable variance over time. Additionally, the skewness of -0.45 suggests a left skew and kurtosis of 2.43 is close to the normal distribution. The AR model provides a reasonable fit with a low AIC of -211.549.

The AR model evaluation metrics summary:

The AR model evaluation metrics provide insight into its forecasting accuracy for the real exchange rate as follows:

- Mean Absolute Error (MAE): The value 0.014244 indicates that, on average, the model's predictions deviate by 0.0142 units from the actual values.
- Mean Squared Error (MSE): The value of 0.000272 represents the average squared difference between predicted and actual values, suggesting a moderate level of accuracy.
- Root Mean Squared Error (RMSE): The value of 0.016503 reflects a typical forecast error of about 0.0165 units, which is close to the MAE, indicating consistency in errors.
- Mean Absolute Percentage Error (MAPE): The value of -13.43% is unusual, as MAPE is typically a positive percentage. A negative MAPE might indicate an incorrect sign handling in the percentage term.

```
                               SARIMAX Results
==============================================================================
Dep. Variable:     Real exchange rate   No. Observations:                   43
Model:                 ARIMA(1, 0, 0)   Log Likelihood                 108.797
Date:                Fri, 04 Apr 2025   AIC                           -211.594
Time:                        08:15:26   BIC                           -206.311
Sample:                    04-01-2014   HQIC                          -209.646
                         - 10-01-2024
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0020      0.003      0.641      0.522      -0.004       0.008
ar.L1         -0.0261      0.154     -0.169      0.866      -0.329       0.277
sigma2         0.0004      0.000      3.607      0.000       0.000       0.001
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):                 2.09
Prob(Q):                              0.98   Prob(JB):                         0.35
Heteroskedasticity (H):               0.73   Skew:                            -0.45
Prob(H) (two-sided):                  0.56   Kurtosis:                         2.42
===================================================================================
```

Figure 20: AR Model Results

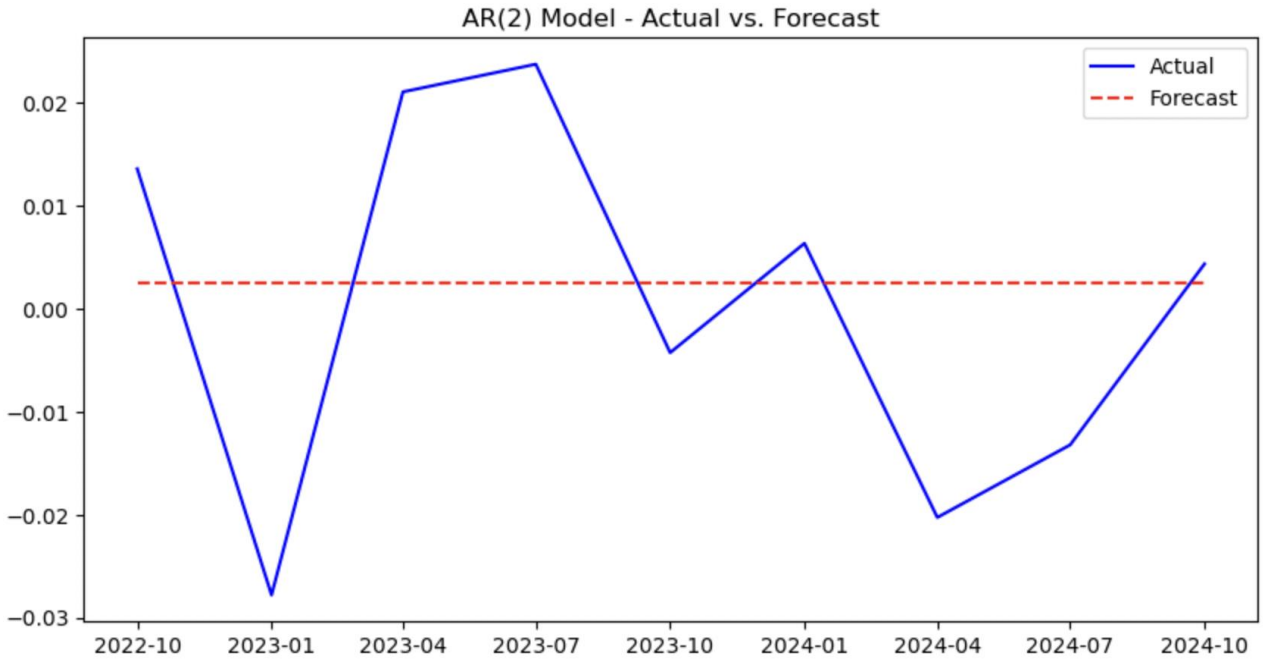

Figure 21: Actual vs. AR (2) Model Forecast

Moving Average model (MA)

Moving Average model is a time series model where the current value is influenced by the past error term or the random shocks rather than the past values. MA model is denoted by (q) (Dubrin, 1986).

Interpretation on ACF plot for MA model

The ACF plot for the Moving Average (MA) model shows a strong autocorrelation at lag 1, which gradually declines and falls within the confidence interval after a few lags. This pattern suggests that the time series follows an MA process, likely of order MA (2), since the correlations quickly taper off beyond the second lag. The sharp cutoff in autocorrelation is a key characteristic of an MA model, differentiating it from an AR model, where the decay would be more gradual. The shaded region represents the confidence interval, and the fact that most points beyond lag 2 lie within it indicates that those lags do not significantly contribute to explaining the time series variation.

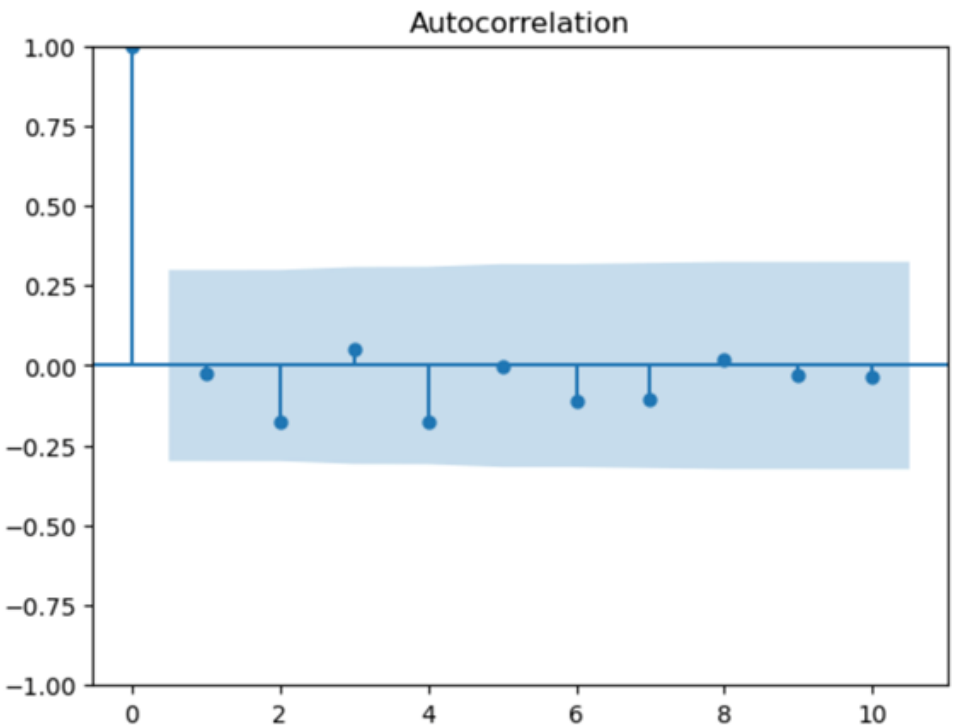

Figure 22: Autocorrelation Function (ACF) Plot

Interpretation on PACF plot for MA model

The Partial Autocorrelation Function (PACF) plot for the Moving Average (MA) model shows a strong negative correlation at lag 1, indicating that the current value is significantly influenced by the immediately preceding observation. Beyond lag 1, the partial autocorrelation values decrease and fall within the confidence interval, confirming the short-term dependency characteristic of an MA process. The sharp cutoff after the first lag suggests that the series follows a low-order MA model, meaning only recent past observations contribute to the time series behaviour.

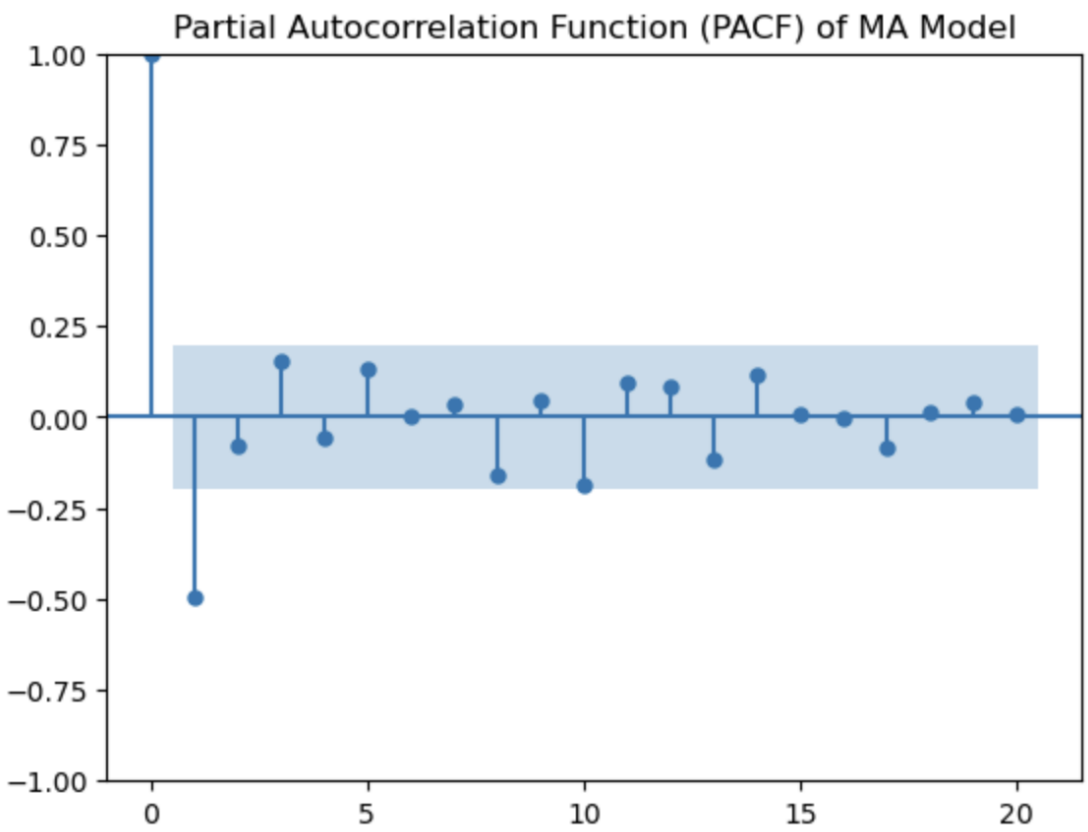

Figure 23: PACF Plot for MA Model

MA model summary

Model diagnostics reveal that the residuals exhibit no significant autocorrelation, as indicated by the Ljung-Box test ($p = 0.95$), suggesting that the model adequately captures the time series' dependencies. The Jarque-Bera test ($p = 0.35$) confirms that the residuals follow a normal distribution, while the heteroskedasticity test ($p = 0.56$) suggests stable variance over time. The slight negative skew (-0.46) and kurtosis (2.41) imply that the residuals are approximately normally distributed but slightly left-skewed.

Overall, while the MA (1) model has a slightly lower AIC of -211.612 compared to the AR (1) model, its key parameters remain statistically insignificant, indicating that it does not provide a strong predictive advantage. Given these results, exploring models combining autoregressive and moving average components (such as ARMA or ARIMA with differencing) may improve the model's forecasting ability.

The MA model evaluation metrics summary

The MA (2) model evaluation metrics suggest its performance in forecasting the real exchange rate.

- Mean Absolute Error (MAE): The value of 0.013888 indicates that, on average, the model's predictions deviate by 0.0139 units from the actual values.
- Mean Squared Error (MSE): The value of 0.000254 shows the squared differences between predicted and actual values, with a lower value implying better accuracy.
- Root Mean Squared Error (RMSE): The value of 0.015926 suggests that the typical deviation from actual values is around 0.0159 units, providing a more interpretable measure of forecast error.

- Mean Absolute Percentage Error (MAPE): 92.92% is significantly high, implying that the model's predictions are often far from actual values. MAPE suggests poor predictive capability, indicating that the model is struggling to capture the patterns in the real exchange rate data effectively.

```
                               SARIMAX Results
==============================================================================
Dep. Variable:     Real exchange rate   No. Observations:                   43
Model:                 ARIMA(0, 0, 1)   Log Likelihood                 108.806
Date:                Wed, 02 Apr 2025   AIC                           -211.612
Time:                        19:52:42   BIC                           -206.328
Sample:                    04-01-2014   HQIC                          -209.664
                         - 10-01-2024
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0020      0.003      0.658      0.511      -0.004       0.008
ma.L1         -0.0435      0.157     -0.278      0.781      -0.351       0.264
sigma2         0.0004      0.000      3.605      0.000       0.000       0.001
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):                 2.10
Prob(Q):                              0.95   Prob(JB):                         0.35
Heteroskedasticity (H):               0.73   Skew:                            -0.46
Prob(H) (two-sided):                  0.56   Kurtosis:                         2.41
```

Figure 25: MA Model Results

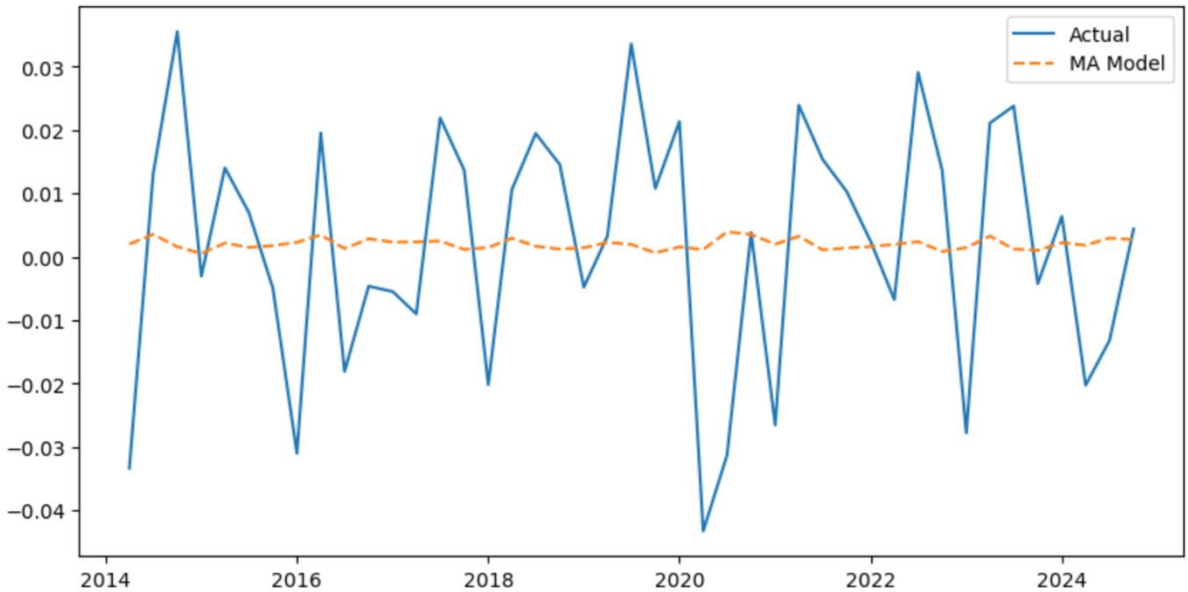

Figure 24: Actual vs MA Model forecast

ARMA model

Autoregressive Moving Average model is a combination of Autoregressive (p) and Moving Average (q), which observes the past values to predict future values and uses past error terms to adjust the predictions (Benjamin, Rigby, & Stasinopoulos, 2003).

Interpretation of ACF and PACF on ARMA model

The graph below consists of the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots for an ARMA (Auto Regressive Moving Average) model. The ACF plot on the left shows a strong correlation at lag 1, followed by a gradual decline, indicating the presence of both autoregressive (AR) and moving average (MA) components in the model. The slow decay suggests that past values significantly influence future values, characteristic of an AR process. On the other hand, the PACF plot on the right shows a sharp drop after lag 1, which indicates an AR component of order 1 or 2. The presence of negative spikes in both plots further supports the influence of an MA component. The combination of these patterns suggests that the time series follows an ARMA (p, q) process, where past values and past forecast errors contribute to the predictions. To determine the precise values of p and q, further model selection techniques such as AIC or BIC can be applied.

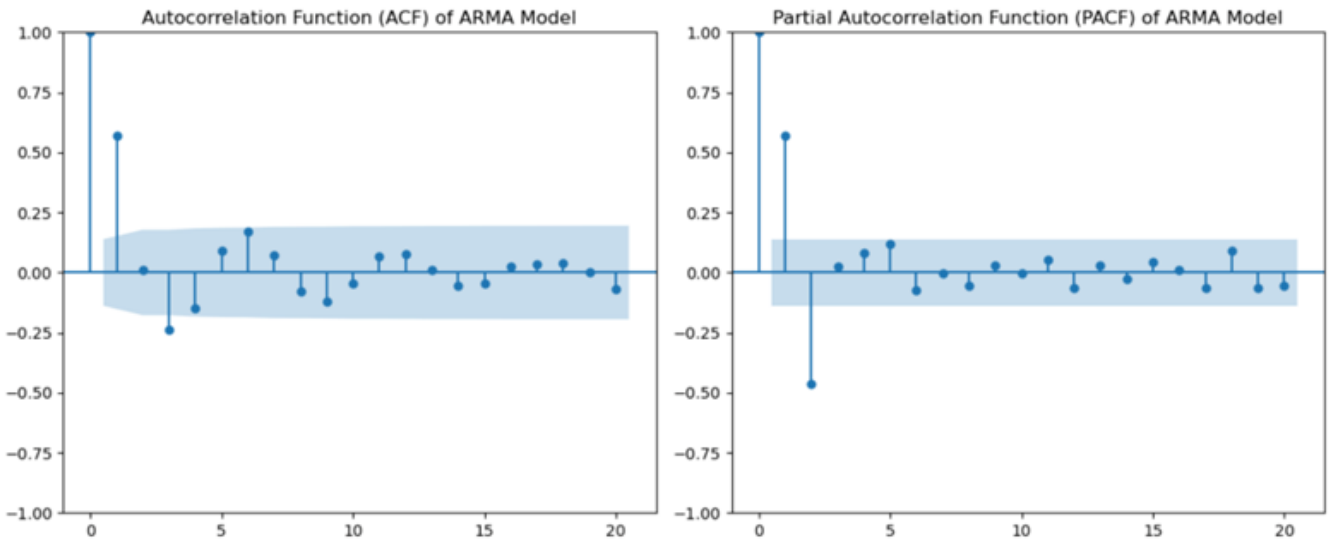

Figure 26: ACF and PACF for ARMA Model

ARMA model summary

Model diagnostics show that the residuals do not exhibit significant autocorrelation, as suggested by the Ljung-Box test (p = 0.49), indicating that the model has adequately captured the underlying dependencies in the data. The Jarque-Bera test (p = 0.24) confirms that the residuals follow a normal distribution, while the heteroskedasticity test (p = 0.41) suggests stable variance over time. The slight negative skew (-0.56) and kurtosis (2.42) indicate a distribution close to normal but with a mild left-skew.

Although the AIC value of -211.344 is slightly higher than that of the MA (1) model, the statistical significance of both AR (1) and MA (1) terms suggests that the ARMA (1,1) model provides a better fit for the data by capturing both past values and past errors.

ARMA evaluation metrics

The evaluation metrics for the ARMA (2,2) model indicate its forecasting accuracy and potential limitations.

- Mean Absolute Error (MAE): The value of 0.015285 suggests that, on average, the model's predictions deviate by approximately 0.0153 units from actual values.
- Mean Squared Error (MSE): The value of 0.000304 indicates the average squared deviation, with smaller values generally implying better predictive accuracy.
- Root Mean Squared Error (RMSE): The value of 0.017441 provides a more interpretable measure of forecast error, showing that typical deviations from actual values are around 0.0174 units.

Mean Absolute Percentage Error (MAPE): The percentage of 100.14% is exceptionally high. This suggests that the model struggles significantly in capturing the real exchange rate data patterns.

```
                               SARIMAX Results
==============================================================================
Dep. Variable:     Real exchange rate   No. Observations:                   43
Model:                 ARIMA(1, 0, 1)   Log Likelihood                 109.672
Date:                Wed, 02 Apr 2025   AIC                           -211.344
Time:                        19:52:42   BIC                           -204.299
Sample:                    04-01-2014   HQIC                          -208.746
                         - 10-01-2024
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0020      0.004      0.560      0.575      -0.005       0.009
ar.L1         -0.7853      0.202     -3.882      0.000      -1.182      -0.389
ma.L1          0.9151      0.159      5.741      0.000       0.603       1.227
sigma2         0.0004      0.000      3.102      0.002       0.000       0.001
===================================================================================
Ljung-Box (L1) (Q):                   0.48   Jarque-Bera (JB):                 2.87
Prob(Q):                              0.49   Prob(JB):                         0.24
Heteroskedasticity (H):               0.64   Skew:                            -0.56
Prob(H) (two-sided):                  0.41   Kurtosis:                         2.42
===================================================================================
```

Figure 27: ARMA (1,1) Model Results

ARIMA Model

ARIMA model is an advanced technique in time series forecasting, which comprises of Autoregressive (p), Integrated (d), and Moving Average (q) (Shumway & Stoffer, 2017).

ACF and PACF interpretation on ARIMA model

The ACF (Autocorrelation Function) and PACF (Partial Autocorrelation Function) plots shown above help identify the appropriate ARIMA model parameters (p, d, q).

ACF: The first lag shows a strong positive correlation (1.0) because a time series is always perfectly correlated with itself. The next few lags have gradually decreasing correlations, suggesting an MA (Moving Average) component in the data. The ACF eventually dies out within the confidence interval (blue shaded area), which suggests the series is stationary or has been differenced to become stationary.

PACF: The first lag shows a strong positive correlation, followed by a sharp drop. A few significant spikes exist beyond lag 1, then it tapers off, suggesting the presence of an AR

(Autoregressive) component in the data. The cutoff after a few lags in PACF is a sign that the series follows an ARIMA (p, d, q) model with a dominant AR component.

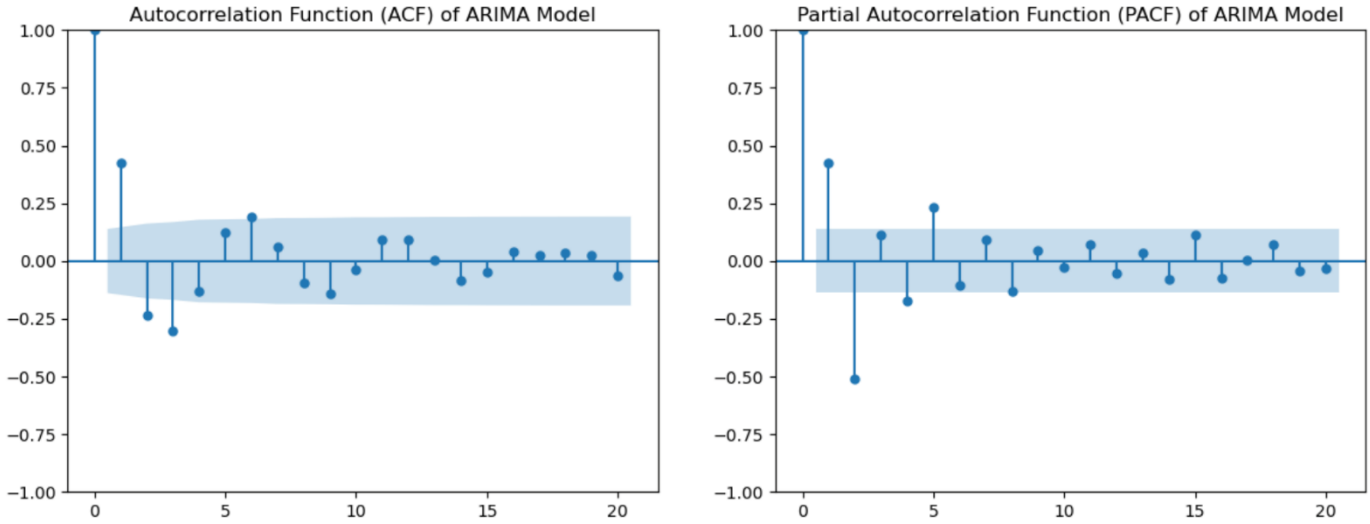

Figure 28: ACF and PACF of ARIMA Model

ARIMA model summary

The ARIMA model evaluation is based on AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion), where lower values indicate better model performance. Among the tested models, ARIMA (0,1,1) has the lowest AIC of -156.64 and BIC of -153.65, making it the best choice regarding fit and complexity. ARIMA (1,1,1) and ARIMA (0,1,2) have slightly higher AIC and BIC values of -154.67 and -150.18, making them reasonable alternatives. ARIMA (2,1,2) and ARIMA (2,1,0) perform worse, with higher AIC and BIC values, while ARIMA (1,1,0) has the highest AIC of -145.31 and BIC of -142.32, indicating the poorest fit. Based on these results, ARIMA (0,1,1) is the most suitable model, with ARIMA (1,1,1) and ARIMA (0,1,2) as backup options.

ARIMA (0,1,1)

The model selection criteria, particularly the AIC of -156.643, indicate that this ARIMA (0,1,1) model provides a good fit to the data compared to other models tested. The residual diagnostics further support the model's reliability. The Ljung-Box test (p = 0.70) suggests no significant autocorrelation in the residuals, indicating that the model captures most of the time-dependent structure in the data. Additionally, the Jarque-Bera test (p = 0.52) confirms that the residuals follow a normal distribution, while the heteroskedasticity test (p = 0.66) suggests stable variance over time.

The skewness (-0.47) and kurtosis (2.75) indicate that the residuals are slightly left-skewed but still relatively normal. Given these results, the ARIMA (0,1,1) model is a strong candidate for forecasting the real exchange rate, as it effectively captures past dependencies while ensuring stationarity.

```
Best Model Summary:
                               SARIMAX Results
==============================================================================
Dep. Variable:     Real exchange rate   No. Observations:                   34
Model:                 ARIMA(0, 1, 1)   Log Likelihood                  80.322
Date:                Thu, 03 Apr 2025   AIC                           -156.643
Time:                        22:03:43   BIC                           -153.650
Sample:                    04-01-2014   HQIC                          -155.636
                         - 07-01-2022
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ma.L1         -0.9783      0.455     -2.152      0.031      -1.869      -0.087
sigma2         0.0004      0.000      1.627      0.104   -8.45e-05       0.001
===================================================================================
Ljung-Box (L1) (Q):                   0.15   Jarque-Bera (JB):                 1.29
Prob(Q):                              0.70   Prob(JB):                         0.52
Heteroskedasticity (H):               1.31   Skew:                            -0.47
Prob(H) (two-sided):                  0.66   Kurtosis:                         2.75
===================================================================================
```

Figure 29: ARIMA (0,1,1) Model Summary

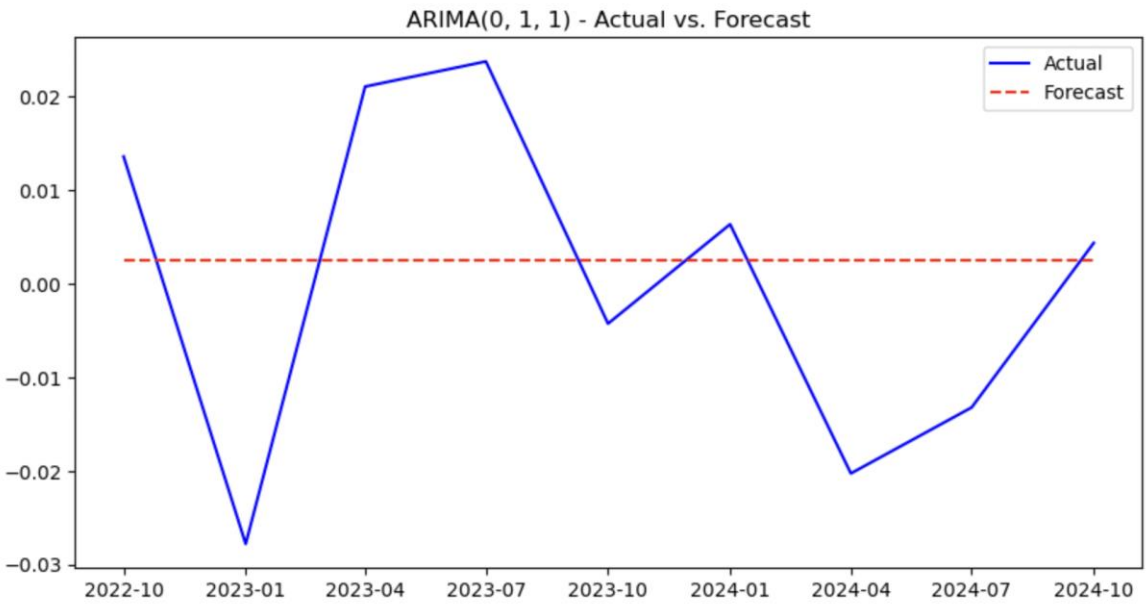

Figure 30: Actual versus ARIMA (0,1,1) Forecast

ARIMA evaluation metrics

The evaluation metrics for the ARIMA model provide insights into its forecasting accuracy are as follows:

- **Mean Absolute Error (MAE): The value** of **0.014691** indicates the average absolute difference between the predicted and actual values, showing that, on average, the forecast deviates by approximately 0.0147 units.
- **Mean Squared Error (MSE):** The value of **0.000296** further highlights the model's performance by penalizing larger errors more heavily, ensuring that predictions do not have extreme deviations.
- **Root Mean Squared Error (RMSE):** The value of **0.017217** suggests that the typical forecasting error is around 0.0172 units, providing an easily interpretable measure of forecast accuracy.
- **Mean Absolute Percentage Error (MAPE):** The value of **95.80%** is notably high, indicating that the model's predictions, on average, deviate significantly from actual values in percentage terms.

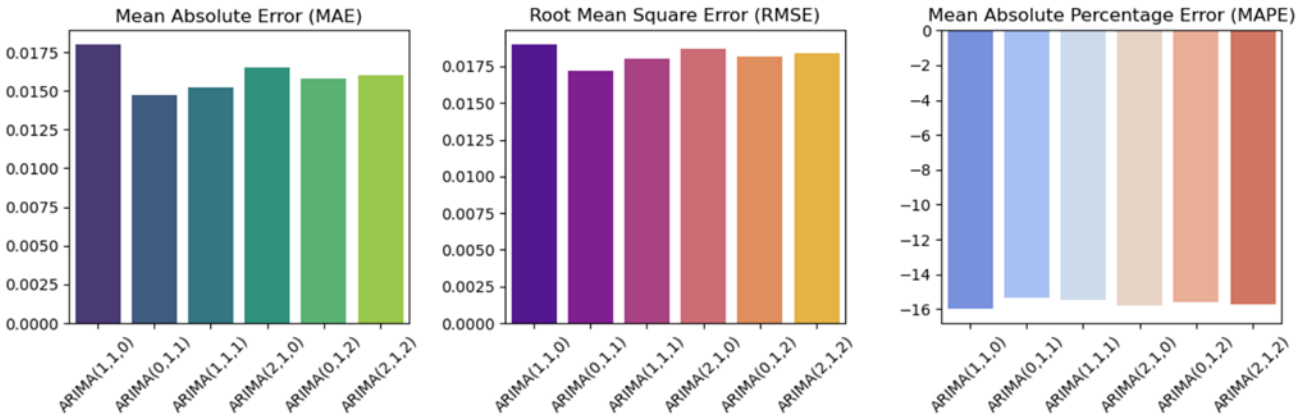

Figure 31: Model Comparison Using Forecast Error Metrics

Arima (0,1,1) Forecast

The ARIMA (0,1,1) forecasting model displays actual data from January to December 2024 while forecasting the upcoming 12 months' values. The blue line shows varying actual past values under zero, but the green line in the model stays at -1.85. The model presents increasing prediction uncertainty through wider shading in the 95% confidence interval area over time. Although the model predicts stable trends ahead, there will be substantial variations in the data points. The flat projection pattern is standard for ARIMA (0,1,1) models since they track short-term variations but do not include long-term patterns.

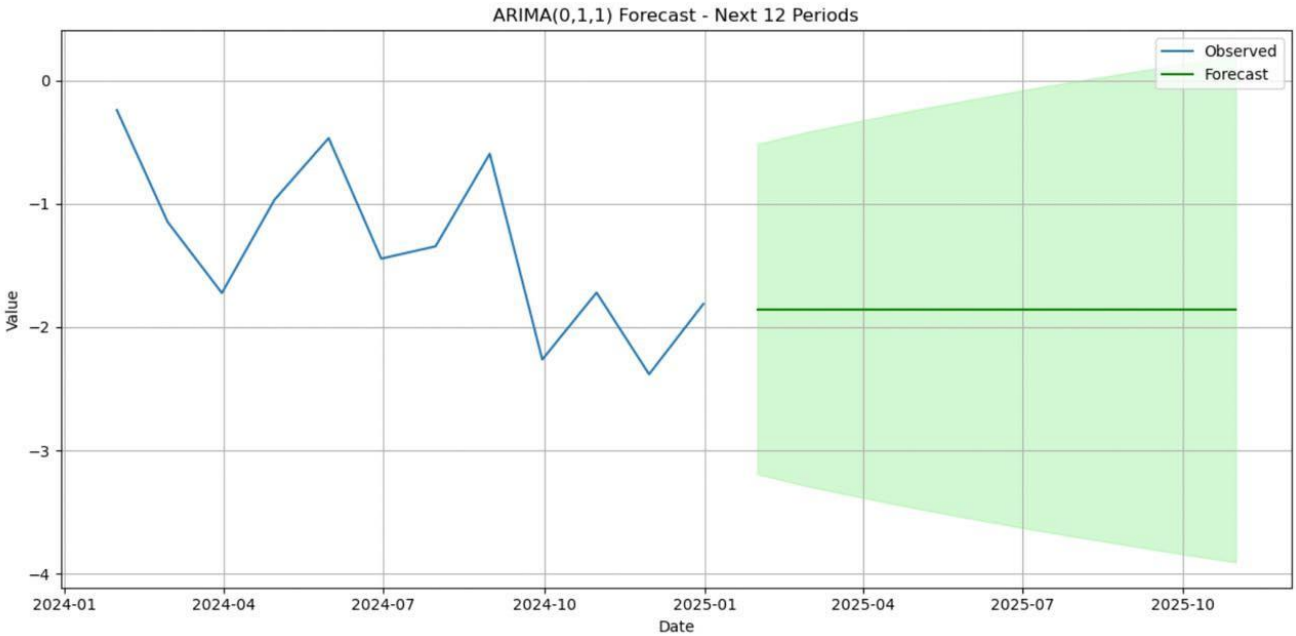

Figure 32: ARIMA (0,1,1) 12-Period Forecast

Conclusion

Applying the Box-Jenkins procedure enabled us to model and forecast the real exchange rate dynamics between Canada and Australia from 2014 to 2024. We developed and compared several time series models by carefully transforming the data and performing extensive stationarity tests. The ARIMA (0,1,1) model emerged as the optimal fit based on selection criteria, even though forecast errors remained relatively high. Our findings imply that factors beyond simple inflation differentials drive exchange rate fluctuations. Overall, this approach sheds light on the complexities of currency behaviour and underscores the challenges inherent in accurately predicting exchange rate movements.

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Appendix

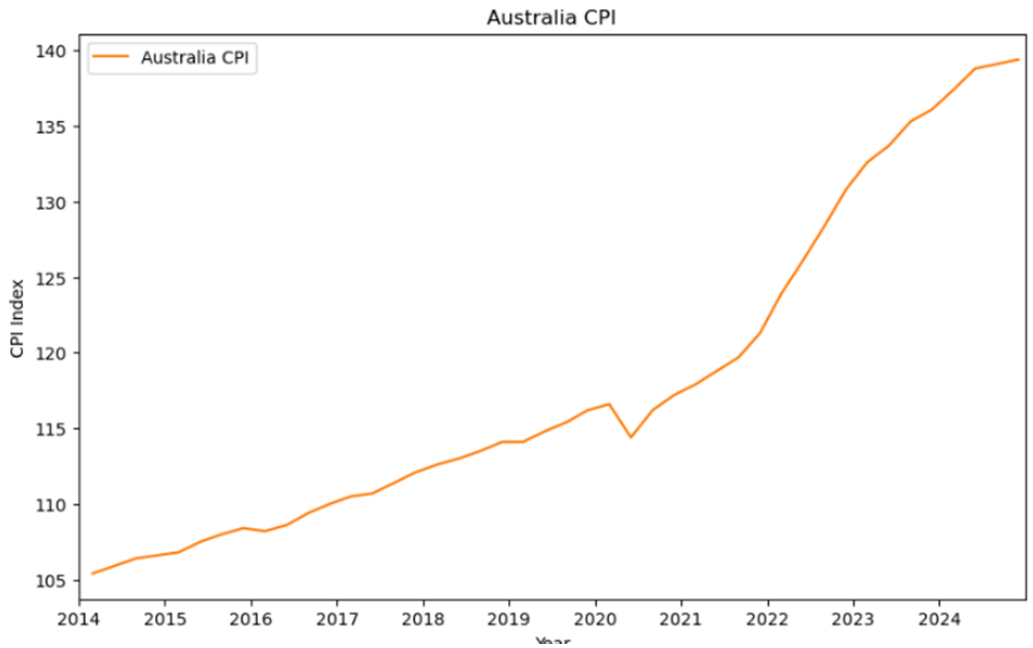

Figure 33: Australia's CPI (2014–2024)

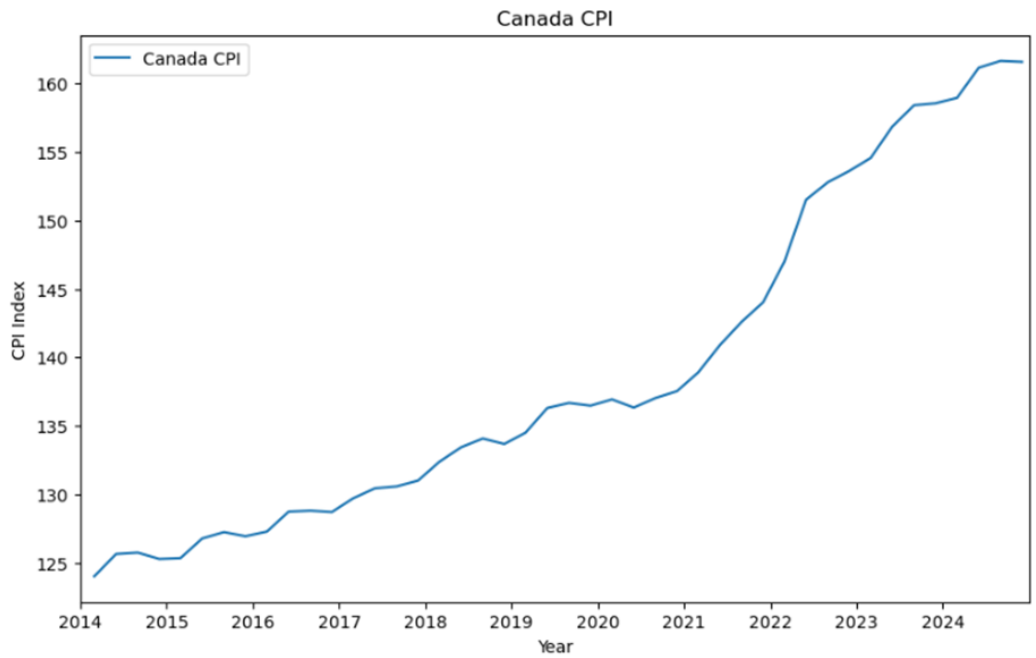

Figure 34: Canada's CPI (2014-2024)